

The Moon Race: USSR vs US

Intelligent Agents — Exercise 03
University of Haifa

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1 Cold War Context and the Space Race Begins

1.1 Superpower rivalry after 1945

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Table 1: Selected Moon Race milestones

Year	Program	Milestone
1957	Sputnik	First artificial satellite
1961	Vostok	First human orbital flight
1965	Gemini	Rendezvous and EVA practice
1969	Apollo 11	First crewed lunar landing
1975	Apollo-Soyuz	Post-race cooperation



Figure 1: Apollo 11 Saturn V on Launch Pad 39A before the Moon landing mission.

סיכום הפרק

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2 Soviet Early Lead: Sputnik and Vostok

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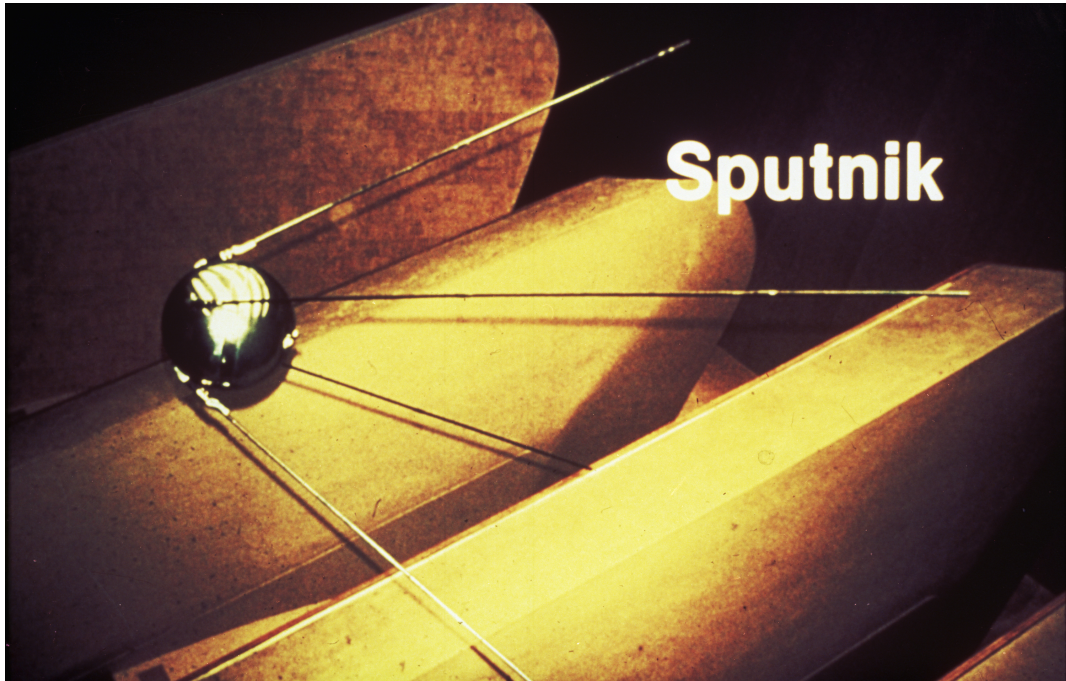


Figure 2: Sputnik era — the Soviet Union launched the first artificial satellite in 1957.

סיכום הפרק

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3 American Response: Mercury, Gemini, Apollo

3.1 NASA and the crash program

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The ideal rocket equation relates velocity change to mass ratio:

$$\Delta v = v_e \ln\left(\frac{m_0}{m_f}\right) \quad (1)$$

where v_e is effective exhaust velocity, m_0 initial mass, and m_f final mass.

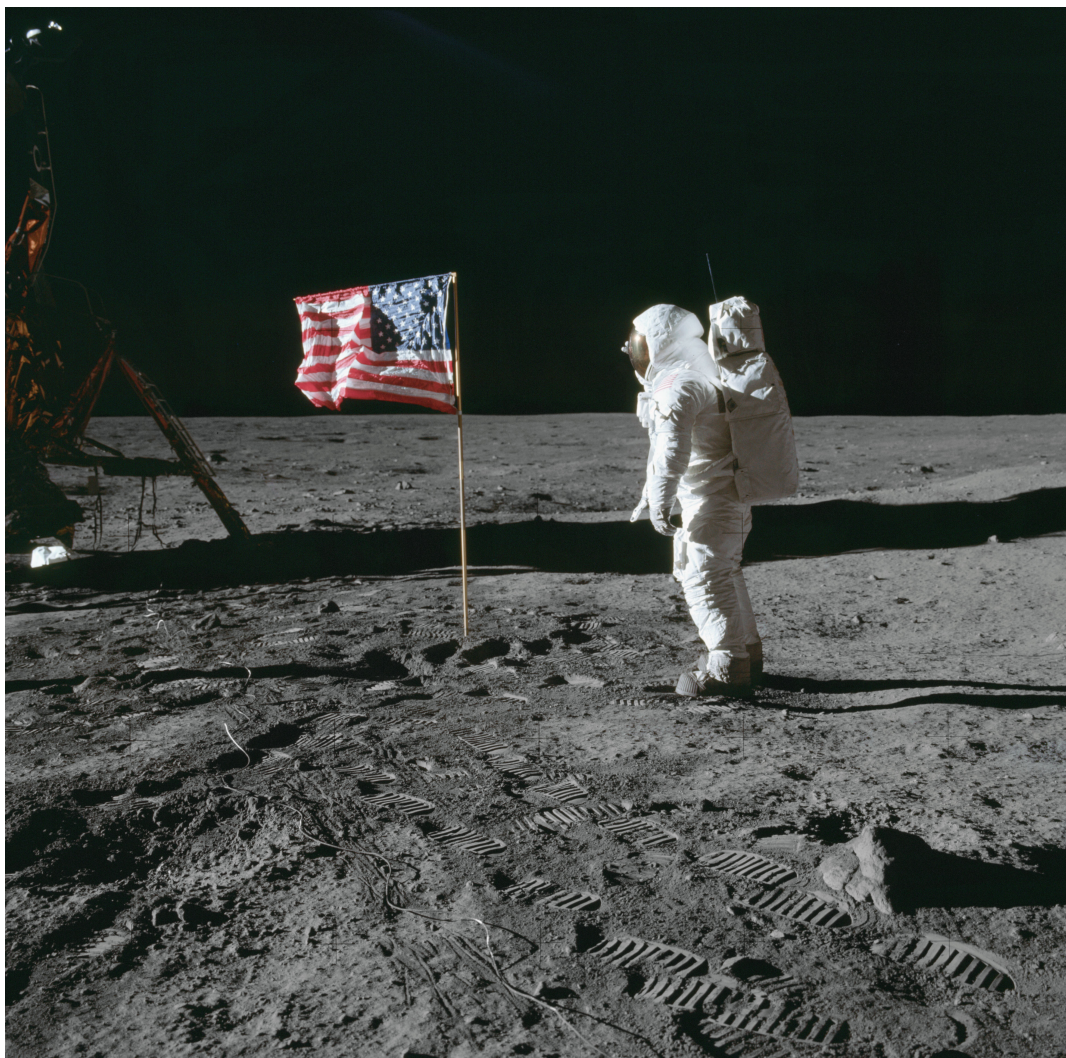


Figure 3: Buzz Aldrin on the Moon during Apollo 11.

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4 Key Missions and Turning Points

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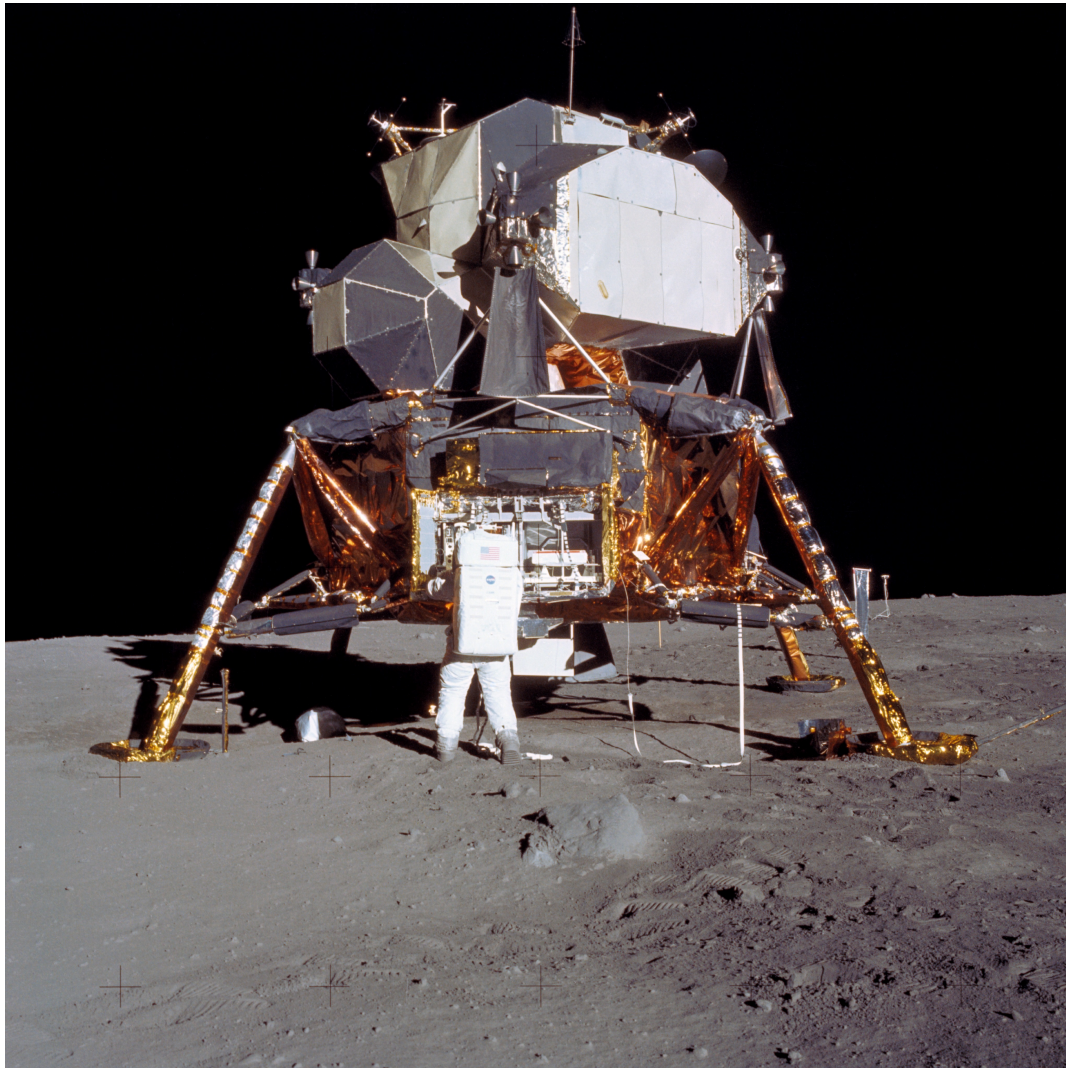


Figure 4: Apollo 11 Lunar Module Eagle on the Moon surface.

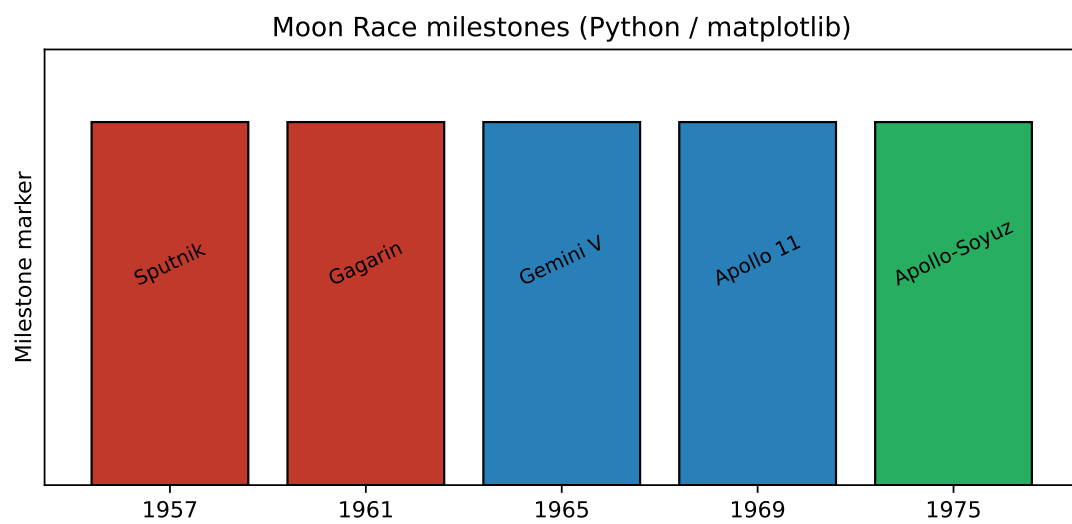


Figure 5: Moon Race milestones chart generated with Python (matplotlib).

סיכום הפרק

פרק זה סוקר משימות מפתח: גשושיות, מיפוי ונחיתה ראשונה של בני אדם על הירח. משימות אלו הפחיתו סיכון והביאו לדגימות מהירח.

5 Propaganda, Politics, and Public Opinion

5.1 Media and ideological messaging

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Figure 6: Earthrise from Apollo 8 — icon of public opinion and the space race.

סיכום הפרק

פרק זה בוחן תעמולה, פוליטיקה ודעת קהל. שני הצדדים הציגו הצלחות חלל כהוכחה לעליונות מערכתית.

6 Legacy: Who Won and What Remains

6.1 End of Apollo and later cooperation

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6.2 Lessons for science and exploration

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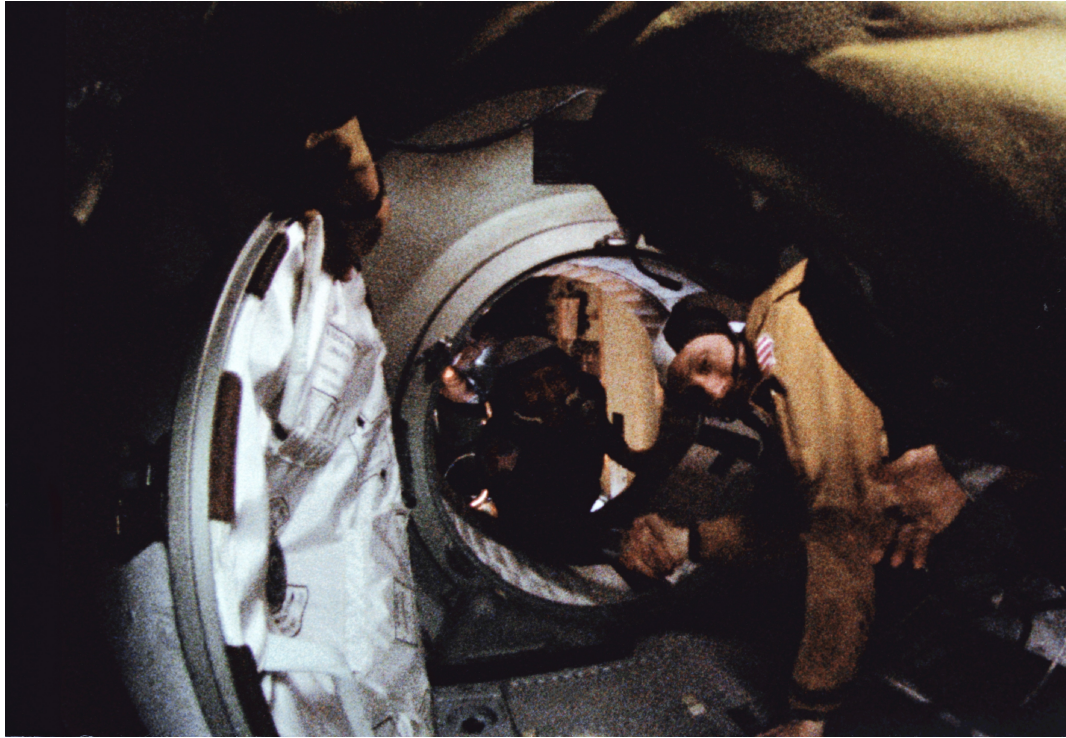


Figure 7: Apollo–Soyuz handshake in space — beginning of post-race cooperation.

סיכום הפרק

פרק זה מסכם את המורשת: סיום תוכניות הנחיתה ושיתוף פעולה בינלאומי. המרוץ הסתיים בייצור ידע ובשיתוף מדעי לאחר שנים של תחרות.

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